

Mohammad Tahsin

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Research Profile

PhD student in Engineering with a Computer Science concentration at Texas A&M University-Kingsville and Graduate Research Assistant in the AI-Based Autonomous System Research Lab. Research spans machine learning for biomedical signals, machine learning applications, explainable AI, cybersecurity, etc.

Research Interests

Biomedical signal analysis and AI for healthcare • Explainable machine learning • Conformal Risk Prediction • Cybersecurity and anomaly detection • Deep Learning

Education

Texas A&M University-Kingsville, Kingsville, TX	Aug. 2026–Present
PhD in Engineering, Computer Science concentration	
Texas A&M University-Kingsville, Kingsville, TX	Jan. 2025–Aug. 2026
MS in Computer Science	
BRAC University, Dhaka, Bangladesh	May 2017–Feb. 2022
BS in Computer Science	

Research Experience

Graduate Research Assistant AI-Based Autonomous System Research Lab	May 2025–Present
Texas A&M University-Kingsville, Kingsville, TX	
- Developed a smart-home sensor placement decision support model with GRU-based energy forecasting; published the work at IEEE ICCE 2026. - Developed a hybrid model for heartbeat monitoring with explainable AI methods. - Investigated deep learning methods for malware detection.	

Current Project

- Zero-Day Intrusion Detection in Consumer Healthcare IoMT with Conformal Rejection (Under review in IEEE Transaction Series)
- AI-Driven Sensor Placement and Wireless Sensor Deployment: A PRISMA-Guided Systematic Review of Optimization Methods, Validation Maturity, and Benchmarking (Under review in MDPI journal)
- CGLGCTNet: A Lightweight Cross-Gated Local Global CNN Transformer Network for Plant Disease Classification (Under review in IEEE journal).
- FetalSANet: A Lightweight, Explainable Spatial-Attention MobileNetV2 Framework for Fetal Ultrasound Plane Classification (Under review in MDPI journal)
- FedQDFU: Non-IID Federated Quantum Learning for Diabetic Foot Ulcer Classification from Deep Image Features (Under review at MDPI journal)
- Data-Local Soft-Voting Ensemble for Intrusion Detection in Consumer IoT Sensor-Gateway Networks (Under review in IEEE conference)
- Explainable Evasive PDF Malware Screening with TabPFN for Consumer Applications (Under review in IEEE conference)
- Systematic Evaluation and Explainability Analysis of Transfer Learning Models for Maize Leaf Disease Classification (Under review in IEEE conference)
- Interpretable Dual-Branch ConvNeXt–Transformer Fusion for Fine-Grained Acute Lymphoblastic Leukemia Classification (Under review in IEEE conference)
- PneuShield-FL: Differentially Private Federated Pneumonia Screening with Deep CXR Features and Explainable AI (Under review in IEEE conference)
- SafeSNM-QFed: A Leakage-Safe Stacked CNN–BFO Framework for Brain Tumor MRI Classification with Quantum and Federated Learning Extensions (Under preparation)
- Energy-Regularized Latent Synthetic Exposure for Open-Set Recognition of Clinical Tabular Data (Under preparation)

Publications

2026

- Mohammad Sadman Tahsin**, H. Y. Adarbah, and A. Noore, “X-Beat: An Explainable Framework for ECG Image Classification,” 2026, arXiv preprint, arXiv:2609.22631
- S. Ahmed, N. Firoz, **Tahsin**, **Mohammad Sadman**, and D. Akpulum, “Forecasting Carbon Footprint: Temporal Fusion Transformer Modeling for Predicting CO2 Emissions in Bangladesh,” in *Innovations in Cybersecurity and Data Science*, S. M. Basha, H. Taherdoost, and N. Bačaniin-Džakula, Eds. Cham: Springer Nature Switzerland, 2026, pp. 387–401, doi: 10.1007/978-3-032-12243-8_26
- Tahsin**, **Mohammad Sadman**, H. Y. Adarbah, and A. Noore, “A Lightweight Channel-Attention CNN for Robust Beat-Level Arrhythmia Detection,” *IEEE Access*, vol. 14, pp. 62 803–62 817, 2026, doi: 10.1109/ACCESS.2026.3686810
- Tahsin**, **Mohammad Sadman**, H. Y. Adarbah, and A. Noore, “Smart-Home Sensor Placement Decision Support via Hybrid Model with GRU Energy Forecasting,” in *2026 IEEE International Conference on Consumer Electronics (ICCE)*, 2026, pp. 1–6, doi: 10.1109/ICCE67443.2026.11449749

2024

- N. Firoz, **Tahsin, Mohammad Sadman**, and M. Z. Hossain, "Autoencoder-Driven Anomaly Detection for Efficient Malware Identification," in *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, 2024, pp. 1–8, doi: 10.1109/ICoICI62503.2024.10696300
- Mohammad Sadman Tahsin**, S. Abdullah, M. A. Karim, M. U. Ahmed, F. Tafannum, and M. Y. Ara, "A comparative study on data mining models for weather forecasting: A case study on Chittagong, Bangladesh," *Natural Hazards Research*, vol. 4, no. 2, pp. 295–303, 2024, doi: 10.1016/j.nhres.2023.12.014

2023

- Tahsin, Mohammad Sadman**, M. Al Karim, M. U. Ahmed, F. Tafannum, and N. Firoz, "An integrated approach for diabetes detection using fisher score feature selection and capsule network," *Journal of Computer Science and Engineering (JCSE)*, vol. 4, no. 2, pp. 61–77, 2023
- G. Cano-Verdugo, J. Owoyemi, A. Khan, **Tahsin, Mohammad Sadman**, D. N. Shukurullaevna *et al.*, "Leveraging Artificial Intelligence to Revolutionize Healthcare: A Comprehensive Analysis," *SciWaveBulletin*, vol. 1, no. 1, pp. 12–19, 2023
- N. Firoz, O. G. Beresteneva, A. S. Vladimirovich, **Tahsin, Mohammad Sadman**, and F. Tafannum, "Automated Text-based Depression Detection using Hybrid ConvLSTM and Bi-LSTM Model," in *2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS)*, 2023, pp. 734–740, doi: 10.1109/ICAIS56108.2023.10073683

2022

- Tahsin, Mohammad Sadman**, N. Firoz, S. A. H. Chowdhury, S. Wasei, M. A. Karim, and M. Y. Ara, "Investigation of Credit Card Fraud detection under the lens of Comparative Machine Learning models," in *2022 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)*, 2022, pp. 25–31, doi: 10.1109/3ICT56508.2022.9990897
- M. Islam, **Tahsin, Mohammad Sadman**, F. Alam, S. S. Hossain, A. Deb, and S. Kar, "Understanding Convolutional Neural Network's behavior for Alzheimer's disease on MRI," in *2022 IEEE 13th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, 2022, pp. 0179–0185, doi: 10.1109/IEMCON56893.2022.9946529
- M. Al Karim, **Tahsin, Mohammad Sadman**, M. U. Ahmed, M. Y. Ara, and S. A. Hasnat Chowdhury, "Evaluating the Performance of ID3 Method to Analyze and Predict Students' Performance in Online Platforms," in *2022 International Conference on Computer Science and Software Engineering (CSASE)*, 2022, pp. 60–64, doi: 10.1109/CSASE51777.2022.9759650
- M. U. Ahmed, M. A. Karim, **Tahsin, Mohammad Sadman**, Y. Rahman, and F. Tafannum, "Analyzing Co2 Emission in Developing Countries Using Auto Regression and Auto Regression Walk Forward: A Time Series Approach," in *2022 IEEE Delhi Section Conference (DELCON)*, 2022, pp. 1–5, doi: 10.1109/DELCON54057.2022.9752807
- Tahsin, Mohammad Sadman**, M. Jobayer, M. B. U. Antor, M. Islam, F. F. Raisa, and M. A. H. Shaikat, "Predictive Analysis & Brief Study of Early-Stage Diabetes Using Multiple Classifier Models," in *2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*, 2022, pp. 0203–0207, doi: 10.1109/CCWC54503.2022.9720736

2021

- Tahsin, Mohammad Sadman**, M. Y. Aziz, T. A. Kabbo, T. Tahsin, N. h. Zumme, and M. I. Hossain, "Data Security Model Using Deep Learning and Edge Computing for Internet of Things (IoT) in Smart City," in *2021 19th OITS International Conference on Information Technology (OCIT)*, 2021, pp. 381–386, doi: 10.1109/OCIT53463.2021.00081
- Tahsin, Mohammad Sadman**, M. A. Karim, M. U. Ahmed, Y. Rahman, F. Tafannum, and S. Abdullah, "Comparative Analysis of Weather Prediction Using Ensemble Learning Models and Neural Network," in *2021 19th OITS International Conference on Information Technology (OCIT)*, 2021, pp. 325–330, doi: 10.1109/OCIT53463.2021.00071

Professional Experience

Credentialing Analyst | Radiant Data Systems Ltd., Dhaka

Mar.–Oct. 2024

- Verified documents and coordinated with U.S. clients to resolve missing information and complete daily operational tasks.

Product Management Intern | REDX, ShopUp, Dhaka

Feb.–June 2022

- Contributed to Google Maps and GPS-based fleet work using real-time traffic information for approximately 100 trucks.

Technical Skills

Programming: Python | ML and data tools: TensorFlow, Keras, scikit-learn, NumPy, pandas, Weka | Methods: explainable AI, time-series analysis, data mining, data visualization

Honors and Awards

- Doctoral Fellowship, Texas A&M University-Kingsville, Fall 2026–Present
- Dean's Graduate Merit Scholarship, Texas A&M University-Kingsville
- Research Assistantship Grant, Texas A&M University-Kingsville, Fall 2025
- Vice Chancellor's List, BRAC University, Summer 2020

Professional Service

Peer reviewer: Heliyon; Biomedical Signal Processing and Control; Intelligence-Based Medicine; Primary Care Diabetes; Systems and Soft Computing; Natural Hazards Research; Journal of Computer Science; Franklin Open; Expert Systems with Application; ICDATE 2023.

Technical Program Committee: IEEE ICCE 2026.